

Vehicle Control System - High Level Design

Prototype Evaluation Document - Synthetic Data

Field	Value
Document ID	HLD-VCS-001
Version	1.0
System	Vehicle Control System
Classification	Synthetic / Evaluation Only

1. Purpose

This High Level Design describes the software architecture of the Vehicle Control System. The document defines software components, ports, interfaces, signals, dependencies and high-level functional flow for prototype evaluation.

2. Software Components

Component	Responsibility
WheelSpeedSensor	Acquires wheel-speed information and provides vehicle speed data.
VehicleSpeedController	Processes wheel-speed information and calculates validated vehicle speed.
InstrumentCluster	Displays validated vehicle speed to the driver.

3. Interfaces

Interface	Provider	Consumer
WheelSpeedInterface	WheelSpeedSensor	VehicleSpeedController
VehicleSpeedInterface	VehicleSpeedController	InstrumentCluster

4. Port Definitions

Component	Port	Port Type	Interface
WheelSpeedSensor	WheelSpeedOut	P-Port	WheelSpeedInterface
VehicleSpeedController	WheelSpeedIn	R-Port	WheelSpeedInterface
VehicleSpeedController	VehicleSpeedOut	P-Port	VehicleSpeedInterface
InstrumentCluster	VehicleSpeedIn	R-Port	VehicleSpeedInterface

5. Signal Definitions

Signal	Interface	Unit
WheelSpeed	WheelSpeedInterface	km/h
VehicleSpeed	VehicleSpeedInterface	km/h

6. Component Dependencies

Dependent Component	Required Component
VehicleSpeedController	WheelSpeedSensor
InstrumentCluster	VehicleSpeedController

7. Functional Flow

WheelSpeedSensor -> VehicleSpeedController -> InstrumentCluster

8. Revision Information

Current document version: 1.0. Document identifier: HLD-VCS-001.